

Bachelor of Science in Civil Engineering

Year of study: 2nd Year; 2nd Semester

Course Title: CIVIL ENGINEERING DRAWING AND
DESIGN

Course Code: FCC3 242

LECTURE ONE

OVERVIEW AND GENERAL INFORMATION

CCED

Lecture Outline

- 1.1 Introduction
- 1.2 Learning Outcome from this lecture
- 1.3 Course objectives and expected outcomes
- 1.4 Course Content
- 1.5 Mode of Delivery
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1.1 Introduction

Welcome to this first lecture in FCC3 242 course. The lecture covers the overview and general information. In this lecture, we are going to discuss the syllabus and its requirements. This will enable you to form a general picture of what the course entails and what is expected of you. In this lecture, we shall start by discussing the objectives and expected learning outcomes, course content, method of delivery, assessment and finally the suggested general reading to supplement lecture notes. The pre-requisite to this course is BAR3 142.

1.2 Learning Outcome from this lecture

At the end of this lecture, you should be able to:

- (a) Have information to encourage you carry out advance reading
- (b) Appreciate the requirements both for the course and yourselves

1.3 FCC3 242 Course objectives and expected outcomes

1.3.1 Course Objectives

This course prepares and gives Civil Engineering student the knowledge and basic ability to visualize and object. It also enhances the skills to present and organize engineering ideas and be able to communicate them graphically.

1.3.2 Expected learning outcomes from FCC3 242 course

At the end of this course, the learner should be able to: -

- a) Utilize different techniques used in graphical communication.
- b) Visualize, present and organize ideas graphically by means of pencil drawings and selected CAD applications
- c) Develop the art of seeing and representing 3 dimensional objects in 2 dimensional views.

1.4 Course Content

The course content is as follows:

- a) *Projections of surfaces and solids*:** Examples of plane surfaces and solids, Projection of an object, Types of projections, Position of the picture plane, Representation of views on drawing paper, Projection of points, lines, planes and solids.
- b) *Section of solids*:** Recap of graphics communication with engineering drawing, Introduction to sectional views, Purposes of section views, Procedure of obtaining section views, Terminologies used in sectional views, cutting plane line styles, Section lining, Kinds of sections, Drawings containing two or more sections, Dimensioning of the section views.
- c) *Development of surfaces*:** Definition of surface developments, Procedure of surface development, Developable and undevelopable surfaces, Types of developments: - Straight-line development, Radial line development of flat surfaces, Radial line development for conical surfaces, Parallel line development of cylindrical surfaces.

Course Content continued

- d) *Intersection of surfaces*:** Examples of engineering objects with intersecting surfaces, Lines and curves of intersection, Establishing lines and curves of intersection in orthographic views of objects with flat and curved surfaces, Developments of intersecting objects.
- e) *Isometric projection*:** Recap of pictorial drawings, uses of pictorial drawings, Types of pictorial drawings, Procedure for producing isometric drawings, Non-isometric lines, Dimensioning Isometric Drawings, Curved Surfaces in Isometric.

1.5 Mode of Delivery

The course unit is delivered as follows:

- a) Two-hour lecture per week – to be held physically

After each topic, the students will be issued with soft copies of lecture notes with worked examples and tutorial questions.

- b) Two-hour tutorial session per week, at the end of each topic – to be held physically

1.6 Assessment

This course is examined as follows:

- a) Continuous assessment test (CAT) (drawing manually) = 20%
- b) 5 No. drawing assignments throughout the semester = 40%

Submission of coursework assignments

All assignments shall be submitted physically on the agreed dates usually two weeks after the assignment have been issued.

- c) 3-hour examination at the end of semester (drawing manually) = 40%

1.7 Suggested Reference Books

- a) Gichaga, F. J. and Rangasami, K. S. (1986) Introduction to Building and Civil Engineering Drawing. ANSTI Technology Series. Low Cost Edition. MacMillan Publishers. London.
- b) Reekie Fraser, (1976). Draughtsmanship. Architectural and Building Graphics. 3rd Edition. Edward Arnold Publishers Ltd. London.
- c) Parker and Pickup. (1981). Engineering Drawing with worked examples. Volume 1 and 2. 3rd Edition. Stanley Thornes Publishing Ltd. Cheltenham. England.
- d) Jensen, C and Helsel, J. D. (1990). Engineering Drawing and Design. 4th Edition. Gregg Division. MacGraw-Hill Publishing Company
- e) Ellen Finkelstein (2010). AutoCAD® 2010 & AutoCAD LT® 2010 Bible. Wiley Publishing Inc.